Theory-Guided Transfer Learning for Macroeconomic Forecasting: Embedding a Real Business Cycle Prior into a Residual Network

Joshua Lim
joshlim0@student.ubc.ca

Abstract

We adapt the theory-guided transfer-learning framework of Chen et al. (2024), originally developed for option pricing, to quarterly macroeconomic forecasting. The core idea is to treat a simple economic simulator as a pretraining dataset: generate millions of synthetic trajectories from it, pretrain a neural network on them, and then fine-tune on the much smaller amount of real economic data. A residual MLP is first pretrained on approximately 12.5 million transition pairs simulated from a calibrated two-shock Real Business Cycle (RBC) model, with a physics-informed-style auxiliary loss that penalises predictions inconsistent with the simulator's equilibrium equations. The pretrained weights are then fine-tuned sequentially on a cross-country macro panel (10 OECD economies, 1980–2019), and then on US-specific financial conditions (1990–2019). We evaluate against AR(1), VAR(2), and Bayesian VAR (BVAR) time-series baselines on a held-out 2014–2019 window, event-study rollouts over the 2001 and 2008 recessions, and an out-of-sample window covering 2023–2025. The results are mixed and interpretable. In-sample, VAR(2) dominates one-step forecasting on all three target variables in the calm 2014–2019 expansion; the pretrained network alone is catastrophically bad (R^2 on the order of -10^3); fine-tuning reduces error by roughly two orders of magnitude but does not close the gap to VAR(2). Out of sample over 2023–2025, however, the picture inverts for the latent investment-efficiency shock $\hat{\xi}$ (a latent state the RBC model uses to absorb whatever investment dynamics it cannot otherwise explain): the fine-tuned network attains $R^2 = +0.174$ where AR(1) collapses to $R^2 = -4.84$, because AR(1)'s training-sample mean is far from the OOS regime. For the productivity shock $\hat{z}$ (the total-factor-productivity, or TFP, disturbance), the *pretrained* network surprisingly beats all fine-tuned variants OOS, suggesting the RBC prior captures TFP dynamics well across cycles. The main contribution is a demonstration that the Chen-et-al. template — theory-as-prior via synthetic pretraining with a structural auxiliary loss — transfers to a small-sample macro setting where it produces gains that are specifically regime-robust rather than universally better, consistent with the structural prior doing exactly the work of anchoring predictions when the data distribution moves.

1 Introduction

Quarterly macroeconomic forecasting is a small-sample problem. US real-activity series contain roughly 250 observations at post-war quarterly frequency, and even cross-country panels of advanced economies rarely exceed ∼2,000 country-quarters after conditioning on covariate availability. The standard structural tools in this area — simulators built from microeconomic first principles, known collectively as Dynamic Stochastic General Equilibrium (DSGE) models, of which the Real Business Cycle (RBC) framework of Kydland and Prescott (1982) is the simplest — are theoretically coherent

Course project for UBC CPSC 440/550, April 2026.

but known to be wrong in specific ways: they omit financial frictions, nominal price rigidities, and non-linear crisis dynamics. Modern deep networks sit at the opposite pole: flexible and data-driven, but prone to overfitting on several-hundred-observation macro series and typically lacking any inductive bias that would anchor them during regime shifts.

Chen et al. (2024) propose a bridge. Their *theory-guided transfer learning* recipe, demonstrated for Black–Scholes option pricing, is to (i) generate a large synthetic dataset from a structural model, (ii) pre-train a neural network on that corpus with an economics-informed auxiliary loss that enforces structural identities (e.g. delta-/vega-matching), and then (iii) fine-tune on real market data. The structural model is used as an informative prior over network weights rather than as a forecaster in its own right. In their setting the approach beats both a plain deep-learning baseline and the Heston stochastic-volatility model, and the gains are largest in small samples and under volatility regime change.

This project asks whether their recipe transfers to a regime where the structural model is a DSGE rather than Black–Scholes, the target is macro dynamics rather than derivative prices, and the real-data sample is orders of magnitude smaller. Macro is a much less hospitable domain: the Black–Scholes misspecification is a single volatility-surface distortion; RBC omits financial frictions, nominal rigidities, heterogeneity, and non-linear propagation all at once. Whether the prior survives fine-tuning at all, and whether it survives in a way that actually improves forecasts, is the empirical question.

Contribution. We build out the Chen et al. pipeline end-to-end for a two-shock RBC model and test it against classical time-series baselines. Three things are new relative to Chen et al. (2024): (i) a multi-stage curriculum where fine-tuning goes from a cross-country panel to a single-country target, with the same input dimensionality throughout so that padding slots are gradually replaced with real features; (ii) a country-by-country curriculum in the first fine-tuning stage that uses the different history lengths as a regularizer; and (iii) an evaluation protocol with structural diagnostics (impulse response functions against the analytical RBC solution) alongside forecast accuracy, which lets us separate "is the prior preserved?" from "does it help?" We report *mixed* results: in-sample VAR(2) wins, but out-of-sample the network shows specific regime-robustness gains on the variables whose long-run dynamics the RBC model captures best. Details in §4.3.

2 Related Work

Theory-guided transfer learning. The closest prior work is Chen et al. (2024), who generate Black–Scholes option data, pretrain a feed-forward network on it with auxiliary losses that match the model's analytical derivatives with respect to the underlying price (delta) and volatility (vega), and then fine-tune on observed option surfaces. Their main finding — that a misspecified theoretical model used as a prior beats both an untrained network and the (also misspecified) Heston model — motivates carrying the recipe over to macro. What's new here relative to their work is the domain shift: a different structural model (RBC vs. Black–Scholes), a much smaller real-data sample ($\sim 2{,}000$ country-quarters vs. daily option surfaces), and a multi-stage fine-tuning curriculum that their two-stage template does not have.

Physics-informed neural networks. The auxiliary-loss component is structurally analogous to Raissi, Perdikaris, and Karniadakis (2019), who penalise residuals of governing PDEs during training. Our economics-informed loss replaces PDE residuals with the RBC capital-accumulation identity and the linear consumption policy function (derived analytically from the steady-state log-linearisation), so violations of equilibrium conditions carry gradient signal in addition to the usual state-prediction MSE. The difference from standard PINNs is that the "physics" here is only approximate — RBC is known to be misspecified — so the auxiliary loss acts as a soft inductive bias rather than as ground truth.

Transfer learning from synthetic sources. Yosinski et al. (2014) and the BERT line (Devlin et al. 2019) are the canonical pretraining-then-fine-tuning references; synthetic-to-real transfer is well studied in vision and robotics, less so in macro-finance. The small-sample regime makes transfer both most attractive (pretraining is the main way past the over-parameterisation gap) and most

fragile (fine-tuning easily overwrites the prior); our low fine-tuning learning rate and aggressive early stopping are responses to that fragility.

Time-series benchmarks in macro. The relevant statistical baselines are AR(1), unrestricted VAR (C. A. Sims 1980), and shrinkage BVARs with Minnesota-style priors (Doan, Litterman, and C. Sims 1984; Bańbura, Giannone, and Reichlin 2010). We compare against all three.

3 Method

3.1 Source domain: a two-shock RBC model

The RBC model is a small economic simulator. At each quarter t it tracks four state variables: capital stock $\hat{k}_t$, consumption $\hat{c}_t$, a productivity disturbance $\hat{z}_t$, and a capital-efficiency (investment) disturbance $\hat{\xi}_t$. The two disturbances are exogenous AR(1) processes that drive the economy away from its steady state; the other two variables respond optimally to them according to the choices of a single idealised agent ("representative household"). We use the standard two-shock formulation of Justiniano, Primiceri, and Tambalotti (2010).

For ML purposes the only detail that matters is that, after log-linearising around the steady state (the deterministic long-run equilibrium of the model), the system reduces to a linear Gaussian state-space transition $s_{t+1} = Ps_t + Q\epsilon_{t+1}$ with a *closed-form* consumption policy $\hat{c}_t = a_k\hat{k}_t + a_z\hat{z}_t + a_\xi\hat{\xi}_t$. Both P and (a_k, a_z, a_ξ) are analytic functions of the structural parameters and recovered via a generalised eigenvalue decomposition; full equations in Section A.1. Having closed-form access to the policy is what makes the economics-informed loss in §3.4 possible. We calibrate the model to US quarterly moments via Simulated Method of Moments (SMM, a standard moment-matching procedure), fixing α=0.33, β=0.99, δ=0.025 at textbook values and estimating the rest.

3.2 Synthetic corpus for pretraining

From the calibrated solver we draw 50,000 parameter vectors independently from uniform priors centred on the SMM estimates, then simulate each draw for 250 quarters after a 500-quarter burn-in. Pairing consecutive periods gives approximately 12.5 million transition examples $(s_t \rightarrow s_{t+1})$. Sampling parameters rather than fixing them forces the network to learn the *map from states to next-period states*, not the trajectory of any one calibration. The ML reading of this is that we are compressing a family of nearby economies into a single pretrained regressor.

Input-dimensionality stability. We pad inputs to a fixed 19 dimensions across all stages (4 state variables + 15 auxiliary slots). During pretraining the 15 slots carry uniform noise (not zeros, which would let the network learn to ignore them and leave cold weights for later fine-tuning to reallocate); the slots are progressively replaced with real features in Stages 1–2.

3.3 Architecture

A residual MLP (He et al. 2016) with the following structure is used: an input projection (Linear → LayerNorm (Ba, Kiros, and Hinton 2016) → GELU) from 19 to hidden dimension 512; four residual blocks at width 512, each applying Linear → LN → GELU → Dropout(0.1) → Linear → LN with an additive skip and post-activation GELU; and a linear output to the three target dimensions $(\hat{k}_{t+1}, \hat{z}_{t+1}, \hat{\xi}_{t+1})$. Total parameter count is ≈ 2.1M. Consumption $\hat{c}_{t+1}$ is recovered analytically from the policy function and is not a network target.

3.4 Economics-informed loss

A plain MSE on the network's three outputs $(\hat{k}', \hat{z}', \hat{\xi}')$ would train the network to match sample paths but would not penalise predictions that are internally inconsistent with the simulator. This is the same failure mode that physics-informed neural networks (Raissi, Perdikaris, and Karniadakis 2019) address for PDEs. Following Chen et al. (2024), we add two auxiliary structural penalties.

The key observation is that the simulator's closed-form policy and identities let us *reconstruct* the quantities the network did not predict: implied next-period consumption $c'_{\text{pred}} = a_k\hat{k}' + a_z\hat{z}' + a_\xi\hat{\xi}'$

comes from the policy function, and implied next-period investment i'_{pred} comes from the goods-market identity applied to the implied output y_{pred} from the production function. These should equal the simulator's realised c_{t+1}, i_{t+1} if the network's state predictions are correct. The auxiliary loss penalises the discrepancy:

$$\mathcal{L} \;=\; \lambda_0 \underbrace{\mathbb{E}\big[|c'_{\text{pred}} - c_{t+1}| + |i'_{\text{pred}} - i_{t+1}|\big]}_{\mathcal{L}_{\text{controls}}\ \text{(policy consistency)}} \;+\; \lambda_1 \underbrace{\mathbb{E}\Big[\big|\hat{k}' - \big((1-\delta)\hat{k}_t + \delta\hat{i}_t + \delta\hat{\xi}_t\big)\big|\Big]}_{\mathcal{L}_{\text{dynamics}}\ \text{(capital accumulation)}}$$

with $\lambda_0 = \lambda_1 = 1$. $\mathcal{L}_{\text{controls}}$ forces the network's predicted states to be consistent with the simulator's optimal choices; $\mathcal{L}_{\text{dynamics}}$ forces them to be consistent with the physical law of motion for capital (next-period capital equals undepreciated current capital plus realized investment). The net effect is an inductive bias that the network's outputs must jointly satisfy the simulator's equilibrium, not just match it pointwise.

3.5 Three-stage fine-tuning curriculum

Stage 0 (Pretrain). AdamW (Loshchilov and Hutter 2019) with lr=10^{-3}, weight decay 10^{-4}, batch 2048, cosine-annealed over 50 epochs (Loshchilov and Hutter 2017). Validation loss falls from 0.147 to 0.0625 (converged).

Stage 1 (Cross-country panel). We fine-tune on a panel of 10 advanced economies (CAN, DNK, FIN, JPN, NLD, NOR, SWE, CHE, GBR, USA), 1980Q1–2019Q4. The four RBC state variables are reconstructed per country from the Penn World Tables (PWT), a standard cross-country macro dataset (Feenstra, Inklaar, and Timmer 2015): annual series are interpolated to quarterly frequency with a cubic spline, then passed through the Hodrick–Prescott filter (Ravn and Uhlig 2002) — a standard low-pass filter in macroeconomics that separates a smooth trend from the business-cycle component. The latent investment-efficiency shock $\hat{\xi}$ has no direct observable counterpart and is instead backed out as the residual of the consumption policy function. Ten additional macro features (GDP growth, unit labour cost growth, CPI inflation, unemployment, long–short yield spread — each as a level and an 8-quarter rolling standard deviation) fill input positions 4–13; positions 14–18 remain zero-pad.

Countries are fine-tuned one at a time, ordered by earliest data availability. For each country, AdamW restarts at lr=10^{-4}, batch 32, with gradient clipping at norm 1.0, cosine annealing, and early stopping on an 80/20 chronological split (patience 10). After each country we reload the best-validation-loss checkpoint before moving on, so countries with longer histories anchor the prior before shorter-history countries are introduced — a curriculum-learning idea (Bengio et al. 2009) applied at the domain level rather than the example level.

Stage 2 (US-specific fine-tune with financial features). We replace the five zero-pad slots with five US financial-conditions features, chosen to proxy dimensions the RBC model omits: the BAA–AAA corporate bond spread (credit risk), the Fed funds rate (monetary policy stance), log VIX (aggregate uncertainty), the 10-year-to-Fed-funds term spread (yield-curve slope), and the S&P 500 quarterly log return (equity-market state). We fine-tune on 120 US quarters (1990Q1–2019Q4) split 95 train / 24 validation, using AdamW with a much smaller lr=10^{-5}, batch 8, cosine annealing over 100 epochs, and patience 20. Because the network already encodes real macro dynamics from Stage 1, Stage 2 is targeted recalibration of the five new input channels rather than general-purpose training.

The whole pipeline keeps identical input dimensions throughout, which is what lets us swap padding slots for real features between stages without architectural change.

4 Experiments

4.1 Setup

Evaluation is on held-out data the network has never seen during its corresponding stage. The in-sample window is the Stage 2 validation split (2014Q1–2019Q4, 24 quarters). The out-of-sample window is 2023Q1–2025Q4 (11 quarters), reconstructed from FRED with the same HP-filtering pipeline used for training; 2020–2022 is deliberately skipped because the COVID shock represents a structural break outside the training distribution and would contaminate the test.

Table 1: One-step-ahead forecast accuracy on the 2014Q1–2019Q4 validation window (24 quarters). Lower MAE/RMSE is better; R^2 higher. Green cells are the overall best per column. Red cells flag the pretrained network's catastrophic in-sample R^2 values, included to show the magnitude of error before any fine-tuning. **Bold** marks the best learning-based (network) method per column.

	$\hat{k}$ (capital)			$\hat{z}$ (TFP)			$\hat{\xi}$ (inv. shock)		
Model	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
AR(1)	0.00030	0.00039	$+0.71$	0.00043	0.00054	$+0.81$	0.0322	0.0542	$+0.50$
VAR(2)	0.00015	0.00021	$+0.93$	0.00021	0.00028	$+0.96$	0.0189	0.0234	$+0.85$
BVAR	0.00032	0.00038	$+0.64$	0.00057	0.00068	$+0.73$	0.0330	0.0410	$+0.52$
Pretrained	0.0431	0.0484	-4980	0.1252	0.1275	-10506	0.0982	0.1099	-1.06
Stage 1	0.00412	0.00513	-55	0.00443	0.00545	-18	0.0925	0.1049	-0.87
Stage 2	**0.00054**	**0.00061**	-0.17	**0.00089**	**0.00101**	$+0.34$	**0.0428**	**0.0615**	$+0.36$

Baselines. Three classical time-series benchmarks. AR(1) fits each target as a first-order autoregression on its own past. VAR(2) (vector autoregression of order 2 (C. A. Sims 1980)) fits $(\hat{k}, \hat{c}, \hat{z}, \hat{\xi})$ jointly on their first two lags with a constant, after standardisation. BVAR is the Bayesian variant of VAR with a Minnesota shrinkage prior (Doan, Litterman, and C. Sims 1984; Bańbura, Giannone, and Reichlin 2010) that pulls coefficients toward a random walk — a standard small-sample regulariser that typically beats unrestricted VAR when the sample size is on the order of the number of coefficients, as it is here. We use λ_1=0.2, λ_2=0.5 for the prior tightness. We also report the pretrained-only network (Stage 0) and the Stage-1-only network as ablations of the fine-tuning curriculum, so the rows form a natural staircase.

Metrics. MAE, RMSE, coefficient of determination R^2, and directional accuracy (fraction of one-step sign matches). We also report structural diagnostics: impulse response functions to a 1σ TFP shock (compared to the analytical P-matrix solution), and long-run autocorrelations of network rollouts (compared to the calibrated ρ_z, ρ_ξ).

4.2 Training diagnostics

Pretraining converges to validation loss 0.0625 by epoch 50, with per-target MAEs on held-out synthetic data of 2.08×10^{-4} for $\hat{k}$ and roughly 1.6×10^{-2} for $\hat{z}$ and $\hat{\xi}$. The much larger MAEs on the shocks reflect a fundamental limit rather than a training failure: shock innovations are white noise, so the network can only learn the deterministic component $\rho \cdot z_t$ and cannot predict the unforecastable ϵ_{t+1}. Rollout autocorrelations and impulse response functions track the analytical solution closely, confirming that the structural prior was successfully installed (Appendix Table 6).

Stage 1 per-country validation losses range from 6.1×10^{-4} (Finland) to 2.0×10^{-2} (United Kingdom), with a median near 3×10^{-3}. Stage 2 converges to validation loss 1.01×10^{-3} at epoch 2, triggering the patience-20 early stop after epoch 22. The extreme earliness of the best epoch indicates that the Stage 1 prior is already close to the Stage 2 optimum on the US subsample — consistent with the transfer hypothesis, but also a caution that Stage 2 is extracting only a thin sliver of additional signal before early stopping.

4.3 Forecast results

Tables 1 and 2 summarise the core forecasting results.

Three patterns emerge.

(1) Fine-tuning is essential. The pretrained network's in-sample R^2 runs to -10^3 and -10^4, and MAE is two orders of magnitude above AR(1). Stage 1 cuts MAE by an order of magnitude on $\hat{k}$, $\hat{z}$ (but not $\hat{\xi}$); Stage 2 cuts another order on all three. Pretraining is an initialiser, not a forecaster.

(2) Classical baselines dominate in-sample. On the calm 2014–2019 window VAR(2) takes the lowest one-step MAE and highest R^2 across *all three* variables, including $\hat{\xi}$. Stage 2 does not beat VAR(2), BVAR, or AR(1) on any in-sample MAE or R^2. The only in-sample metric where Stage 2

Table 2: Out-of-sample forecast metrics, 2023Q1–2025Q4 (skipping the COVID structural break, 11 quarters). Green cells are the overall best per column — note that the winner is a different model for each target variable. Red cells flag the deepest negative R^2 values: AR(1) collapses on $\hat{\xi}$ because the post-COVID starting state is far from its 1990–2013 training-sample mean, while the networks accumulate error on the now-very-smooth $\hat{k}$. The OOS pipeline compared only against AR(1); VAR(2) and BVAR were not refit for the OOS window.

	$\hat{k}$ (capital)			$\hat{z}$ (TFP)			$\hat{\xi}$ (inv. shock)		
Model	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
AR(1)	0.00107	0.00119	$+0.34$	0.00404	0.00468	-0.77	0.0534	0.0980	-4.84
Pretrained	0.00219	0.00238	-1.61	0.00233	0.00301	$+0.27$	0.0348	0.0396	$+0.05$
Stage 1	0.00274	0.00296	-3.04	0.00277	0.00338	$+0.07$	0.0324	0.0400	$+0.03$
Stage 2	0.00251	0.00272	-2.42	0.00276	0.00348	$+0.02$	0.0289	0.0369	$+0.17$

leads AR(1) is directional accuracy on $\hat{\xi}$ (0.87 vs 0.79; DirAcc for VAR/BVAR was not computed). High-persistence linear models suffice in a smooth expansion.

(3) The picture inverts out of sample for $\hat{\xi}$ and $\hat{z}$. On $\hat{\xi}$, Stage 2 attains $R^2 = +0.174$ while AR(1) collapses to -4.84 and Stage 2's MAE (0.0289) is roughly half AR(1)'s (0.0534); the AR(1) failure is a regime-boundary effect, with predictions reverting toward a 1990–2013 mean far from the post-COVID starting state. The network, conditioned on contemporaneous financial features, has no such failure mode. On $\hat{z}$ OOS the *pretrained* network (never shown real data) beats every fine-tuned variant and AR(1), suggesting the RBC prior captures TFP dynamics across cycles and that fine-tuning slightly over-fits to the 2014–2019 window. AR(1) still wins $\hat{k}$ OOS (capital is nearly a random walk). VAR(2) and BVAR were not refit for the OOS window.

Structural validity. Impulse responses to a 1σ TFP shock match the analytical P-matrix solution almost exactly for the pretrained network. Fine-tuning introduces visible deviations, mostly on $\hat{\xi}$ and $\hat{c}$ — the variables that pick up the empirical financial-conditions signal that the RBC model itself does not contain. Rollouts still mean-revert and autocorrelations stay close to the calibrated ρ_z, ρ_ξ across all stages, so the prior is partly eroded but not destroyed.

Event-study rollouts. Rolling forward from 2007Q1 with actual observed financial features, Stage 2 captures the 2008 collapse in $\hat{\xi}$ around quarters 8–10 while Stage 1 and the pretrained rollouts remain flat. The Stage-2 vs. Stage-1 gap is concentrated precisely in the quarters when the BAA–AAA spread and VIX peak — evidence that the five added financial features carry meaningful signal rather than overfitting on 95 quarters. The 2001 dotcom episode is less well captured by all models, consistent with that downturn being equity- rather than credit-driven.

5 Discussion

Strengths. A clean implementation of the Chen et al. (2024) template in a new domain, with a multi-stage curriculum that (to our knowledge) has not been tried for macro forecasting. The result is mixed but interpretable: the network does not beat classical baselines in-sample, but shows specific regime-robustness gains OOS on $\hat{\xi}$ (positive R^2 where AR(1) collapses) and on $\hat{z}$ (the pretrained prior beats AR(1) outright). This is the signature of a structural prior working as advertised: robustness under distribution shift rather than across-the-board accuracy gains.

Weaknesses. The real-data sample is tiny — Stage 2 has 95 training quarters for a 2.1M-parameter network and converges at epoch 2, and the 11-quarter OOS window is too short for proper confidence-interval inference. The embedded RBC is also linearised and omits nominal rigidities; a Smets–Wouters-style medium-scale DSGE (Smets and Wouters 2007) would give a richer prior.

Future work. Three immediate extensions: (i) bootstrap confidence intervals on the OOS deltas, since the 11-quarter sample makes point estimates untrustworthy; (ii) pretrain on a richer DSGE

(medium-scale with nominal rigidities) to test whether the "best prior" is the one closest to truth or the one that leaves more for empirical learning; (iii) replace the chronological country ordering in Stage 1 with a learned ordering by similarity to the target economy — a clean small-scale instance of curriculum learning as a domain-adaptation problem rather than an example-selection one.

Source code for all three stages, the evaluation suite, and the synthetic-data pipeline is at `https://github.com/joshlim01/theory-guided-tl-macro`.

References

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton (2016). "Layer Normalization." *arXiv:1607.06450*.

Marta Bańbura, Domenico Giannone, and Lucrezia Reichlin (2010). "Large Bayesian Vector Auto Regressions." *Journal of Applied Econometrics* 25.1, pp. 71–92.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston (2009). "Curriculum Learning." *International Conference on Machine Learning (ICML)*.

Hui Chen, Yuxiang Cheng, Yueran Liu, and Ke Tang (2024). "Teaching Economics to the Machines." *Working Paper*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." *NAACL*.

Thomas Doan, Robert Litterman, and Christopher Sims (1984). "Forecasting and Conditional Projection Using Realistic Prior Distributions." *Econometric Reviews* 3.1, pp. 1–100.

Robert C. Feenstra, Robert Inklaar, and Marcel P. Timmer (2015). "The Next Generation of the Penn World Table." *American Economic Review* 105.10, pp. 3150–3182.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun (2016). "Deep Residual Learning for Image Recognition." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Alejandro Justiniano, Giorgio E. Primiceri, and Andrea Tambalotti (2010). "Investment Shocks and Business Cycles." *Journal of Monetary Economics* 57.2, pp. 132–145.

Finn E. Kydland and Edward C. Prescott (1982). "Time to Build and Aggregate Fluctuations." *Econometrica* 50.6, pp. 1345–1370.

Ilya Loshchilov and Frank Hutter (2017). "SGDR: Stochastic Gradient Descent with Warm Restarts." *International Conference on Learning Representations (ICLR)*.

– (2019). "Decoupled Weight Decay Regularization." *International Conference on Learning Representations (ICLR)*.

Maziar Raissi, Paris Perdikaris, and George E. Karniadakis (2019). "Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations." *Journal of Computational Physics* 378, pp. 686–707.

Morten O. Ravn and Harald Uhlig (2002). "On Adjusting the Hodrick–Prescott Filter for the Frequency of Observations." *Review of Economics and Statistics* 84.2, pp. 371–376.

Christopher A. Sims (1980). *Macroeconomics and Reality*. Vol. 48, No. 1, pp. 1–48. Econometrica.

Frank Smets and Rafael Wouters (2007). "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach." *American Economic Review* 97.3, pp. 586–606.

Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson (2014). "How Transferable Are Features in Deep Neural Networks?" *Advances in Neural Information Processing Systems (NeurIPS)*.

A Supplementary Material

A.1 RBC state-space equations

After log-linearising the two-shock RBC model around its deterministic steady state, the system reduces to

$$s_{t+1} = P\, s_t + Q\, \epsilon_{t+1}, \qquad s_t = (\hat{k}_t,\, \hat{c}_t,\, \hat{z}_t,\, \hat{\xi}_t), \qquad \epsilon_{t+1} \sim \mathcal{N}(0, \Sigma),$$

with closed-form consumption policy $\hat{c}_t = a_k \hat{k}_t + a_z \hat{z}_t + a_\xi \hat{\xi}_t$. Both P and (a_k, a_z, a_ξ) are analytic in the structural parameters (preferences β, capital share α, depreciation δ, labour elasticity ϕ, shock persistences ρ_z, ρ_ξ) and recovered numerically by a generalised eigenvalue decomposition that keeps the stable manifold.

A.2 Loss reconstruction equations

Given the network's three outputs $(\hat{k}', \hat{z}', \hat{\xi}')$, the auxiliary loss in §3.4 requires reconstructing the model's implied next-period consumption, output, and investment. The full identities are:

$$c'_{\text{pred}} = a_k \hat{k}' + a_z \hat{z}' + a_\xi \hat{\xi}' \qquad \text{(consumption policy function)}$$

$$y_{\text{pred}} = \frac{(\phi+1)\,\hat{z}' + \alpha(\phi+1)\,\hat{k}' - (1-\alpha)\,c'_{\text{pred}}}{\phi+\alpha} \qquad \text{(log-linearised production function)}$$

$$i'_{\text{pred}} = \frac{1}{1-c/y}\,y_{\text{pred}} - \frac{c/y}{1-c/y}\,c'_{\text{pred}} \qquad \text{(goods-market identity)}$$

where (a_k, a_z, a_ξ) are the policy coefficients recovered from the RBC solver for each simulation's parameter vector, ϕ is the inverse Frisch elasticity of labour supply, α is the capital share, and c/y is the steady-state consumption-to-output ratio (calibrated per simulation from its parameters). The third equation inverts the resource constraint $\hat{y}_t = (c/y)\,\hat{c}_t + (1 - c/y)\,\hat{i}_t$.

A.3 SMM calibration

The fixed parameters (α, β, δ) are set at standard quarterly-RBC values; the remaining parameters $(\phi, \rho_z, \rho_\xi, \sigma_z, \sigma_\xi)$ are estimated by Simulated Method of Moments against eight US empirical moments computed from FRED data 1960Q1–2019Q4 (real GDP / consumption / investment growth-rate moments and HP-filtered output autocorrelation). The steady-state consumption ratio c/y is set to the empirical sample mean.

Table 3: RBC parameters: fixed values from the quarterly literature (top block) and SMM-estimated values (bottom block).

Parameter	Definition	Value
α	capital share in production	0.3300
β	household discount rate	0.9900
δ	quarterly capital depreciation rate	0.0250
c/y	steady-state consumption ratio (sample mean)	0.6435
ϕ	inverse Frisch elasticity of labour supply	1.2016
ρ_z	TFP shock persistence	0.7285
ρ_ξ	investment shock persistence	0.8558
σ_z	TFP shock volatility	0.0057
σ_ξ	investment shock volatility	0.1000

Table 4: SMM moment fit. Empirical moments are computed from FRED 1960Q1–2019Q4; model moments are computed from a long simulation at the estimated parameters. g^x denotes the quarterly log-difference of variable $x \in \{y, c, i\}$; "y-cycle" is the HP-filtered output cycle.

Moment	Data	Model	Difference
$\mathbb{E}[c/y]$	0.6435	0.6435	0.0000
std(g^y)	0.0105	0.0094	−0.0011
std(g^c)	0.0103	0.0089	−0.0013
std(g^i)	0.0405	0.0363	−0.0042
std(g^c)/std(g^y)	0.9760	0.9484	−0.0276
std(g^i)/std(g^y)	3.8455	3.8522	+0.0067
acf(y-cycle)	0.7721	0.8354	+0.0633
acf(g^i)	0.1335	−0.0792	−0.2126

The largest discrepancy is on the autocorrelation of investment growth — the model predicts -0.079 against an empirical value of $+0.134$. This is a known limitation of frictionless RBC models in matching short-horizon investment dynamics, and is part of the motivation for the financial-features fine-tune in Stage 2.

A.4 Synthetic-data parameter sampling

Each of the $50{,}000$ synthetic simulations draws an independent parameter vector from the uniform priors below, then simulates the calibrated RBC solver for 250 quarters after a 500-quarter burn-in. The intervals are centred on the SMM estimates (Table 3) with widths chosen to explore the local neighbourhood of the empirically supported region while keeping all draws inside the model's stable-solution manifold.

Table 5: Sampling ranges for the $50{,}000$ synthetic RBC simulations.

Parameter	Calibrated value	Sampling range
α	0.3300	$[0.33 \cdot 0.90,\ 0.33 \cdot 1.10]$
ρ_z	0.7285	$[0.7285 \cdot 0.95,\ \min(0.999,\ 0.7285 \cdot 1.05)]$
ρ_ξ	0.8558	$[0.85,\ 0.95]$
ϕ	1.2016	$[1.0,\ 1.5]$
c/y	0.6435	$[0.6435 \cdot 0.95,\ \min(0.95,\ 0.6435 \cdot 1.05)]$
β	0.9900	$[0.985,\ 0.995]$
δ	0.0250	$[0.020,\ 0.030]$
σ_z	0.0057	$[0.010,\ 0.030]$
σ_ξ	0.1000	$[0.010,\ 0.030]$

A.5 Multi-step rollouts

Rolling the network forward 1, 4, and 8 quarters from Stage 2 validation starts: for $\hat{k}$ and $\hat{z}$ both fine-tuned networks maintain roughly flat error profiles (MAE near 0.005 at $h{=}8$), while the pretrained network degrades slowly from ≈ 0.035 to ≈ 0.040 on $\hat{k}$. For $\hat{\xi}$, the fine-tuned networks' MAE *grows* from ≈ 0.02 at $h{=}1$ to ≈ 0.17 at $h{=}8$ while the pretrained network stays nearly flat at ≈ 0.10. The "structural prior prevents runaway multi-step error" claim thus holds for the RBC-well-described states but not for $\hat{\xi}$, where fine-tuning introduces rollout drift that the structural prior was suppressing.

A.6 Structural diagnostics

Table 6: Structural-diagnostic check on the pretrained network. Autocorrelations from a 2000-step stochastic rollout; calibrated values are the mean ρ in the synthetic prior. IRF errors are ℓ_∞ deviations between network and analytical impulse responses over the first 20 quarters following a 1σ TFP shock.

Diagnostic	Calibrated / analytical	Network	Pass?
ρ_z	0.730	0.731	✓
ρ_ξ	0.860	0.881	✓
$\mathrm{Var}(z)$	2.14×10^{-4}	2.28×10^{-4}	✓
$\mathrm{Var}(\xi)$	3.84×10^{-4}	4.43×10^{-4}	✓
$\mathrm{IRF}_{\hat{k}}\ \ell_\infty$	$\cdot$	3×10^{-5}	✓
$\mathrm{IRF}_{\hat{z}}\ \ell_\infty$	$\cdot$	7×10^{-5}	✓

A.7 Per-country Stage 1 results

The Stage 1 fine-tuning is run sequentially, with country ordering by earliest data availability. Per-country best validation losses are: CAN 3.4×10^{-3} (ep 50), CHE 1.2×10^{-3} (ep 4), DNK 2.1×10^{-3} (ep 5), FIN 6.1×10^{-4} (ep 24), GBR 2.0×10^{-2} (ep 6), JPN 7.1×10^{-3} (ep 6), NLD 5.3×10^{-3} (ep 2), NOR 8.7×10^{-3} (ep 1), SWE 8.6×10^{-4} (ep 22), USA 7.2×10^{-4} (ep 11). The heterogeneity in best-epoch counts is informative: countries where the current prior is already close to optimal (NOR, NLD) plateau immediately, while countries whose dynamics add the most new signal (CAN, SWE) continue to improve for > 20 epochs. GBR's high best-loss reflects the high-inflation 1980s at the start of its sample.

A.8 Event-study rollouts

Omitted from the main body as a figure for space. The full output of the event-study evaluation script plots, for each of the three network stages, 16-quarter rollouts starting from 2000Q1 (dotcom) and 2007Q1 (financial crisis) for $\hat{k}$, $\hat{z}$, and $\hat{\xi}$, with actuals overlaid. The Stage 2-vs-Stage 1 gap on $\hat{\xi}$ is concentrated in the 2008Q3–2009Q1 window, which is exactly when the BAA–AAA spread and the VIX peaked. Stage 1 produces a nearly flat rollout for $\hat{\xi}$ across both episodes.

A.9 Hyperparameters, verbatim.

Pretrain: AdamW (lr $= 10^{-3}$, wd 10^{-4}, cosine-anneal, batch 2048, 50 epochs, seed 42). Stage 1: AdamW (lr $= 10^{-4}$, wd 10^{-4}, cosine-anneal, grad-clip 1.0, batch 32, max 50 epochs per country, patience 10, 80/20 chronological split). Stage 2: AdamW (lr $= 10^{-5}$, wd 10^{-4}, cosine-anneal, grad-clip 1.0, batch 8, max 100 epochs, patience 20). Architecture $19 \to 512 \to 3$ residual MLP, 4 blocks, dropout 0.1, GELU, LayerNorm, ~ 2.1M parameters.